

Enterprise Infrastructure Migration Readiness Report

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Scope and Current State

The infrastructure program is preparing to migrate 62 internal applications from two legacy virtualization clusters to a new private-cloud environment. As of July 31, 26 applications have completed technical discovery, 18 have completed performance testing, and 9 have been migrated to the staging environment. The program covers finance, HR, customer support, engineering tools, and several internal APIs. Production migration has not yet started for the finance systems.

Performance Baseline

Initial tests show that median API response time improved by 19% in the new environment, while storage throughput increased by 31% for the document-management workload. Two older applications showed memory consumption between 22% and 27% higher than baseline. Engineering believes the increase is related to legacy caching behavior, but the root cause is not yet confirmed.

Governance

The steering committee approved an additional three-week performance-validation phase before any Tier-1 application is moved to production. The validation window begins August 19. A requested \$95,000 purchase of an additional observability module remains pending and cannot be ordered until the architecture board completes its review on September 4.

Page 2 - Application Classification and Test Results

Risk Classification

Applications are grouped into Tier 1, Tier 2, and Tier 3. Tier 1 contains 11 systems whose outage would interrupt revenue, payroll, regulatory reporting, or core customer service. Tier 2 contains 23 important internal systems with limited manual workarounds. Tier 3 contains 28 lower-risk applications. The current migration order prioritizes Tier 3 and selected Tier 2 systems while Tier 1 remains blocked pending validation.

Test Results

Of the 18 applications that completed performance testing, 14 met or exceeded the baseline on all critical metrics. Two applications met response-time targets but exceeded memory thresholds. One reporting service failed a 90-minute sustained-load test because of connection-pool exhaustion. One batch-processing service completed successfully but exceeded its overnight processing window by 17 minutes.

Technical Actions

Engineering approved configuration changes for the connection pool and scheduled a retest for August 23. The batch-processing service will receive an additional optimization sprint, but no production date has been approved. The two memory-heavy applications will undergo profiling before a decision is made on whether to refactor or isolate them on larger instances.

Page 3 - Capacity and Security Review

Capacity Forecast

Current private-cloud capacity can support the planned migration with an estimated 34% compute headroom and 28% storage headroom after all 62 applications are moved. This estimate assumes that average utilization remains within the observed Q2 range. If the analytics team proceeds with its proposed real-time data project, compute headroom could fall to approximately 18%. That analytics project is not part of the approved migration scope.

Security Findings

The security team completed configuration reviews on 12 staging workloads. Three medium-severity issues were found: an overly broad service account, an outdated TLS configuration on an internal endpoint, and a storage bucket with inherited permissions that were wider than intended. No critical vulnerabilities were identified. Remediation is required before those workloads can enter production.

Backup and Recovery

A disaster-recovery simulation restored five representative applications within the four-hour recovery-time objective. However, one database required 4 hours and 26 minutes because the backup index had to be rebuilt. The backup team approved a revised indexing process and will repeat the recovery test on September 10. The recovery-time objective itself has not been changed.

Page 4 - Migration Decision and Next Steps

Go/No-Go Conditions

Production migration may begin only when four conditions are met: all Tier-1 performance tests pass; medium-severity security findings are remediated; the repeated recovery test meets the four-hour objective; and application owners approve production cutover plans. Meeting three of the four conditions is not sufficient for Tier-1 systems.

Approved Schedule

The steering committee approved a limited production wave for up to six Tier-3 applications between September 16 and September 20, provided the August validation phase closes without new critical findings. Tier-1 production migration remains unapproved. The program target is to complete at least 40 of the 62 migrations by December 15, with the remaining systems scheduled for Q1 2027.

Management Summary

The migration program is technically promising but not ready for unrestricted production deployment. Performance gains are measurable, yet memory behavior, one load-test failure, and the recovery-test miss require follow-up. The next formal go/no-go review is scheduled for September 12. No change has been made to the approved overall program budget of \$1.8 million.